

JEE MAIN REVISION MASTER: S-BLOCK ELEMENTS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Alkali & Alkaline Earth Metals: Trends & Compounds	8 PYQs	s-Block Chemistry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Alkali & Alkaline Earth Metals: Trends & Compounds	$\text{Plaster of Paris: } \text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$, $\text{Gypsum: } \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, $\text{Heating above } 393 \text{ K} \rightarrow \text{Dead Burnt Plaster}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Alkali & Alkaline Earth Metals: Trends & Compounds (High Yield — 8 PYQs)

Core Principle: Group 1 & 2 trends: Flame colors: Li (crimson), Na (yellow), K (violet), Ca (brick red), Sr (crimson), Ba (apple green). Diagonal relationship of Li with Mg, and Be with Al due to similar charge/radius ratio. Solubility of alkaline earth metal sulphates decreases down the group ($\text{BeSO}_4 > \text{MgSO}_4 > \text{CaSO}_4 > \text{BaSO}_4$) because hydration energy decreases faster than lattice energy. Plaster of Paris: $\text{CaSO}_4 \cdot (1/2)\text{H}_2\text{O}$.

■ **Caution / JEE Pitfall:** Heating gypsum above 393 K (120°C) forms completely anhydrous dead burnt plaster (CaSO_4) which loses the setting property with water!

■ **Speed Trick:** Alkali metals dissolve in liquid ammonia to give deep blue conducting solutions due to ammoniated electrons; in concentrated solution, the color turns bronze and becomes diamagnetic.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following nitrate will decompose to give NO_2 on heating ?..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "The property of the alkaline earth metals that increases with their atomic number is..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "The oxide of potassium that does not exist is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Which one of the following nitrate will decompose to give NO_2 on heating ?

- (A) NaNO_3
- (B) KNO_3
- (C) RbNO_3
- (D) LiNO_3

Correct Answer: None

Solution Analysis: Among the given nitrates, only LiNO_3 will decompose to give NO_2 when heated. In contrast, the nitrates of sodium (Na), potassium (K), and rubidium (Rb) decompose to release oxygen instead. The decomposition reactions are as follows: For lithium nitrate (LiNO_3): $2\text{LiNO}_3 \xrightarrow{\Delta} 2\text{Li}_2\text{O} + 4\text{NO}_2 + \text{O}_2$ For sodium, potassium, and rubidium nitrates (MNO_3), where $\text{M}=\text{Na}, \text{K}, \text{Rb}$): $2\text{MNO}_3 \xrightarrow{\Delta} 2\text{MNO}_2 + \text{O}_2$ Here, LiNO_3 uniquely produces NO_2 along with the metal oxide and oxygen.